Task 12: Day-4 Design Challenge — Re-evaluate & Adapt Design

Operating Systems (CSA04) — Assignment 1 (CO1)

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Marks: 8 Marks (Bloom's: K5 - Evaluate)
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> ***Task 12 Changed Requirement:*** *Add remote patient monitoring at home sending readings over the internet, and migrate patient records to a shared cloud service across partner hospitals. Complete Impact Analysis, Evaluation of Original Design, Adaptation Plan, and Evidence-Based Defense.*

12.A IMPACT ANALYSIS (2 MARKS)

Introducing remote home monitoring and cloud-hosted records changes key operational constraints:

- **Home Monitoring Devices:** Must transmit over untrusted, high-jitter consumer internet connections (Wi-Fi, cellular) rather than an isolated hospital LAN. The OS must handle network dropouts, provide local non-volatile store-and-forward caching, and support secure over-the-air (OTA) updates.
- **Cloud Records Service:** Replaces the local bare-metal cluster with multi-tenant cloud infrastructure. The OS and hypervisor must protect against cross-tenant side-channel attacks, enforce end-to-end encryption in transit and at rest, and adhere to regional data residency regulations.

12.B EVALUATION OF ORIGINAL DESIGN (2 MARKS)

- **Bedside/Home Monitors:** The QNX microkernel architecture *is retained* for local hard real-time sensor processing and immediate alarms (< 10ms). However, the network stack requires redesign: plain LAN protocols cannot be exposed to the public internet.
- **Central Database:** The local two-node Pacemaker cluster *cannot support* multi-facility cloud operations. It lacks WAN-friendly distributed consensus mechanisms (such as Raft or Paxos) and suffers split-brain risks or high lock contention over wide-area latencies.

12.C ADAPTATION AND REDESIGN PLAN (2 MARKS)

The system is adapted to a **Hybrid Edge-Cloud Architecture**:

1. **Edge Home Monitors (Embedded RTOS + Secure Proxy):** The microkernel handles local sensor acquisition and immediate audio/visual alarms. A sandboxed networking daemon buffers telemetry in encrypted flash during outages and streams data via TLS 1.3 mTLS over MQTT/WebSockets when connected.
2. **Cloud-Native Distributed Database:** Transition the data store to a distributed multi-region database (e.g., CockroachDB / Amazon Aurora) deployed across hardened Linux containers. Quorums are coordinated across the three hospitals using the Raft consensus protocol.

12.D EVIDENCE-BASED DEFENSE AND TRADE-OFFS (2 MARKS)

Pillar	Original Local Architecture	Adapted Edge-Cloud Architecture	Defended Trade-Off Justification
Latency	Local LAN: < 1 ms	WAN to Cloud: 20–80 ms; Local Alarm: < 10 ms	Decoupling immediate local alarms from cloud telemetry maintains the 10ms safety deadline regardless of internet conditions.
Availability	99.999% within the hospital	99.999% cloud uptime; 100% edge standalone survival	Store-and-forward flash buffering keeps home monitoring active even during residential ISP outages.
Security	Perimeter firewall defense	Zero-Trust, mTLS, TPM Secure Boot, Enclave Decryption	Protects remote endpoints and federated records against physical tampering, MITM attacks, and multi-tenant data leaks.